Ethical Responsibilities of Engineers and Platforms in Prioritizing User Security

Balancing Cost and Service – A Case Study on Telegram

Course Details

Course Information

Course Name: Engineering Ethics

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Section: 1

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Introduction

Topic Overview:

We are focusing on Cybersecurity and Privacy Ethics in messaging platforms, with Telegram as a case study. We will delve into the engineer's role, the balance between cost and service, and the broader societal implications of these decisions.

Importance of Ethics in IT:

Engineers design digital systems used by billions. Ethical lapses such as poor data protection or misleading designs can cause large-scale harm.

Global and Ethical Relevance:

- Rise of surveillance, data misuse, misinformation, and platform abuse.
- Platforms like Telegram are used for activism, but also for scams and extremist content.
- Balancing openness and security is an ethical tightrope.

Problem Statement

> The rapid evolution of digital platforms like Telegram presents complex ethical dilemmas for various **stakeholders**, including **engineers, users, regulators, and society**. These dilemmas primarily involve balancing robust user security and privacy against operational costs, content moderation, and regulatory demands.

Privacy vs. Societal Security

Balancing user anonymity and data privacy with the need to prevent illicit activities and ensure public safety.

Transparency & Moderation Responsibilities

How platforms communicate policies and enforce content rules for users, while addressing governmental and societal expectations.

Security Investment vs. Operational Cost

The financial implications for the platform in implementing high-level security features, balancing user benefit against economic viability.

Research Objectives

Our study aims to dissect the ethical challenges faced by engineers and platforms in real-world scenarios, particularly within the context of Telegram, to propose actionable ethical guidelines.

Understand Ethical Challenges

Analyze the core dilemmas in prioritizing user security against other factors.

Explore Real-World Cases

Investigate specific instances within Telegram where ethical trade-offs are evident.

Propose Ethical Guidelines

Develop practical recommendations for engineers and platform developers.

Literature Review

Our review synthesized insights from key academic papers to form a comprehensive understanding of the ethical landscape.

Research Summary:

- Telegram provides strong features like Secret Chats, but they are not default.
- Studies show users are misinformed and most never switch to secure mode.
- Lack of transparency and decentralized moderation create risks.

Ethical Guidelines:

- IEEE: Prioritize public safety, disclose limitations.
- ACM: Avoid harm, respect privacy, be honest.
- NSPE: Hold paramount the welfare of the public.

Key Ethical Concepts:

- Informed Consent: Are users aware of encryption levels?
- Duty of Care: What are engineers' obligations?
- Privacy vs. Innovation: Where should lines be drawn?

Methodology

Our research employed a structured approach, integrating theoretical ethical frameworks with practical case study analysis.

1. **Research Approach**
   Extensive review of academic papers on privacy, security, and platform ethics.

2. **Ethical Frameworks**
   Application of Deontology (duty-based ethics) and Utilitarianism (consequence-based ethics) to analyze dilemmas.

3. **Case Study Analysis**
   In-depth examination of Telegram's features and controversies through an ethical lens. It is chosen due to widespread use, unique security model, and documented misuse cases.

Case Study: Telegram Use During the 2019–2020 Hong Kong Protests

Summary of Events:

During the pro-democracy protests in Hong Kong, Telegram became a critical tool for organizing marches, sharing real-time updates, and avoiding government surveillance. Protesters relied on its large group channels and perceived anonymity. The app's decentralized nature and resistance to censorship made it preferable to alternatives like WhatsApp or Facebook.

Ethical Conflicts:

False Sense of Security

Many assumed all chats were encrypted, but default and group chats weren't E2EE, leaving users exposed. Some arrests were linked to leaked data and metadata.

Opaque Governance

Telegram lacks transparency on data handling and abuse reporting, leaving users uncertain about protections.

Minimal Moderation

Loose oversight allowed free expression but also enabled misinformation and doxxing of protesters.

Case Study: Telegram Use During the 2019–2020 Hong Kong Protests

Ethical Analysis:

From a Kantian ethics perspective, Telegram failed to treat users as ends in themselves by not adequately informing them of encryption limitations, denying them the ability to make informed decisions.

From a Utilitarian view, the platform enabled both social good (mobilizing protests) and harm (exposing users), revealing a trade-off between maximizing liberty and minimizing risk.

Engineers and platform designers had an ethical duty to clarify security features and make safer defaults, particularly when serving at-risk populations in politically volatile contexts.

Ethical Issues & Analysis in Telegram

Our analysis reveals several critical ethical issues stemming from Telegram's design and operational choices, particularly concerning user data and platform responsibility.

Category	Ethical Issue	Engineer’s Role	Ethical Framework
Confusing Settings	Privacy	• Design interfaces that clearly communicate privacy levels. • Ensure default settings prioritize user privacy and security.	Deontology: Duty to protect user privacy and avoid misleading design. Utilitarianism: Maximize overall user privacy and trust.
Cost vs. Safety	Security	• Advocate for investment in robust security measures. • Implement end-to-end encryption by default for all communications.	Deontology: Obligation to provide secure communication, irrespective of cost. Utilitarianism: Weighing the benefit of enhanced security against financial implications.

Ethical Issues & Analysis in Telegram (Contd.)

Further exploration of key ethical concerns, highlighting the engineer's responsibilities and ethical frameworks for remaining categories.

Category	Ethical Issue	Engineer's Role	Ethical Framework
Not Enough Information	Fairness / Transparency	• Clearly document and communicate data handling practices and moderation policies. • Establish transparent procedures for responding to government requests.	Deontology: Duty to be truthful and transparent with users about data and moderation. Utilitarianism: Promoting user trust and informed decision-making through transparency.
Control Problems	Fairness / Accountability	• Collaborate with legal and policy teams to address content misuse responsibly. • Develop ethical guidelines for platform governance that balance user rights with societal safety.	Deontology: Responsibility to prevent platform misuse, even with decentralization. Utilitarianism: Minimize harm from illegal content while upholding user freedom.

Telegram's Ethical Trade-offs

Continuing our analysis, Telegram's design choices reveal inherent tensions between desired features and their ethical implications for user security and platform governance.

Convenience & Multi-Device Access	Compromised Default Privacy (Cloud Storage)
Scalability & Large Group Chats	Challenges in Content Moderation
Free Speech & Censorship Resistance	Potential for Misuse & Illegal Content
Rapid Feature Deployment	Overlooked Ethical Design Reviews

Solutions & Recommendations

Addressing these ethical challenges requires a multi-faceted approach involving policy changes, enhanced transparency, and a renewed commitment to ethical design principles.

Default Stronger Privacy

Implement default end-to-end encryption for all communications.

Transparency Guidelines

Publish clear reports on data requests and content moderation.

Improved Moderation

Develop and enforce proactive content moderation policies with community feedback systems.

Integrate Ethics in Design

Establish an ethical review board for new features, prioritizing user well-being.

Role of Engineers

Engineers, particularly in platforms like Telegram, bear significant ethical duties in shaping digital security and user trust.

Core Responsibilities

Ensure user safety, robust data privacy, and a secure system design, safeguarding against vulnerabilities.

Adhering to Ethical Codes

Apply principles from IEEE, ACM, and NSPE to guide decision-making, ensuring responsible technology development.

Balancing Innovation & Trust

Innovate new features while diligently protecting public safety and maintaining unwavering user confidence.

Challenges in Implementation

Despite clear recommendations, implementing ethical security measures faces significant hurdles, from resource constraints to rapidly evolving threats.

Implementation Barriers

Resource limits, internal resistance, and fragmented global privacy policies.

Conflicting Priorities

Tension between profit-driven decisions and ethical responsibilities.

Evolving Threats

Emerging risks like AI misuse, Deep Fakes, SIM swapping, social engineering. and sophisticated cybersecurity attacks.

Future Directions

Looking ahead, the landscape of digital ethics demands greater integration into technology development and governance.

Ethical Governance

Growing role of ethics in platform design and oversight.

Interdisciplinary Collaboration

Encourage efforts between tech, law, and ethics fields.

Key Focus Areas

Prioritize ethical AI, robust cybersecurity, and user data rights.

Conclusion

Dilemmas are Manageable

Ethical dilemmas persist but are manageable with careful consideration.

Engineer's Ethical Balance

Engineers must ethically balance security, cost, and service delivery.

Trust through Transparency

Transparency and strong defaults are crucial for building user trust.

Sustainable Practices

Long-term ethical practices are key to sustainable digital platforms.

References

Our research is grounded in the following key academic works and ethical guidelines:

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Thank You!

The stage is now open for your questions.